FEMM Heat Flow File Formats

Input (`.feh`) and Output (`.anh`) Specifications

D. Meeker

September 13, 2026

Contents

1 Overview

The FEMM heat flow solver (`hsolv`) reads a problem definition from a `.feh` file and writes the solution to a `.anh` file. The `.anh` file begins with a verbatim copy of the input `.feh` file, followed by a `[Solution]` section containing the finite element mesh and computed temperature values.

All floating-point values are written with 17 significant digits (`%.17g` format). Fields within a line are separated by tabs or spaces. Section headers use square brackets (`[Tag]`); property block delimiters use angle brackets (`<BeginBlock>` / `<EndBlock>`). String comparisons during parsing are case-insensitive.

2 The `.feh` Input File

2.1 Global Problem Parameters

Each parameter occupies one line in the format `[Tag] = value`. All are optional; defaults are applied if missing.

Tag	Type	Default	Description
`[Format]`	int	1	File format version
`[Frequency]`	int	0	Always 0 (legacy field)
`[Precision]`	double	1e-8	Nonlinear convergence tolerance
`[MinAngle]`	double	30	Minimum mesh element angle (degrees)
`[DoSmartMesh]`	int	1	Adaptive meshing (0=off, 1=on)
`[Depth]`	double	1.0	Extrusion depth (planar problems)
`[LengthUnits]`	string	millimeters	inches, millimeters, centimeters, meters, mils, microns
`[ProblemType]`	string	planar	planar or axisymmetric
`[Coordinates]`	string	cartesian	cartesian or polar
`[Comment]`	string	(empty)	Problem description (quoted; `\n` = newline)
`[ACSolver]`	int	0	Always 0 (legacy field)
`[dT]`	double	0	Time step for transient analysis (0 = steady-state)
`[PrevSoln]`	string	(empty)	Path to previous solution for transient initialization (quoted)

The `[dT]` and `[PrevSoln]` fields enable transient thermal analysis. When `dT` > 0, the solver performs a single time step of duration `dT`, using the temperature field from `PrevSoln` as the initial condition. Steady-state analysis corresponds to `dT` $= 0$.

2.1.1 Axisymmetric External Region

Written only when `ProblemType` is axisymmetric and both `extRo` and `extRi` are nonzero:

Tag	Type	Description

[extZo]	double	Outer Z coordinate
[extRo]	double	Outer radius
[extRi]	double	Inner radius

2.2 Point (Nodal) Properties

```
[PointProps] = <count>
<BeginPoint>
  <PointName> = "name"
  <Tp> = 0      prescribed temperature (K)
  <qp> = 0      point heat source (W)
<EndPoint>
... repeated count times
```

2.3 Boundary Properties

```
[BdryProps] = <count>
<BeginBdry>
  <BdryName> = "name"
  <BdryType> = 0
  <Tset> = 0        fixed temperature (K)
  <qs> = 0          prescribed heat flux (W/m^2)
  <Tinf> = 0        ambient temperature for convection (K)
  <h> = 0           convection coefficient (W/m^2-K)
  <beta> = 0        radiation coefficient (W/m^2-K^4)
  <TinfRad> = 0     radiation reference temperature (K)
<EndBdry>
```

Boundary types:

- 0 = fixed temperature (Dirichlet): $T = T_{\text{set}}$
- 1 = prescribed heat flux (Neumann): $\hat{n} \cdot F = q_s$
- 2 = convection: $\hat{n} \cdot F = h(T - T_\infty) + q_s$
- 3 = radiation + convection: $\hat{n} \cdot F = h(T - T_\infty) + \beta(T^4 - T_{\infty,\text{rad}}^4) + q_s$
- 4 = periodic
- 5 = antiperiodic

If `TinfRad` is not present in the file, it defaults to the value of `Tinf` (backward compatibility).

2.4 Material (Block) Properties

```
[BlockProps] = <count>
<BeginBlock>
  <BlockName> = "name"
  <Kx> = 1        thermal conductivity, x-direction (W/m-K)
  <Ky> = 1        thermal conductivity, y-direction (W/m-K)
  <Kt> = 0        volumetric heat capacity (J/m^3-K)
  <qv> = 0        volumetric heat generation (W/m^3)
  <TKPoints> = N
    T_0  K_0       temperature-conductivity data (K, W/m-K)
    T_1  K_1       ... repeated N times
    ...
```

```
<EndBlock>
```

When `TKPoints` > 0, the thermal conductivity is nonlinear: $K(T)$ is piecewise-linear interpolated from the T-K curve. Below the first data point, the first conductivity value is used; above the last data point, the last value is used. The solver iterates until convergence when nonlinear materials or radiation BCs are present.

2.5 Conductor Properties

Conductors enforce equipotential (isothermal) constraints or prescribe total heat flow on surfaces, analogous to circuits in the magnetics solver.

```
[ConductorProps] = <count>
<BeginConductor>
  <ConductorName> = "name"
  <Tc> = 0          prescribed temperature (K)
  <qc> = 0          prescribed total heat flow (W)
  <ConductorType> = 0
<EndConductor>
```

Conductor types:

- 0 = prescribed heat flow ($q = q_c$; solver computes T)
- 1 = prescribed temperature ($T = T_c$; solver computes q)

2.6 Geometry

2.6.1 Nodes

```
[NumPoints] = <count>
x    y    propIdx   inGroup   conductorIdx   [lfs]
```

- `x`, `y`: coordinates (double)
- `propIdx`: index into point property list (1-based; 0 = none)
- `inGroup`: group number (int)
- `conductorIdx`: index into conductor list (1-based; 0 = none)
- `lfs`: local feature size constraint (double, optional; -1 = auto)

2.6.2 Line Segments

```
[NumSegments] = <count>
n0    n1    maxSideLength   propIdx   hidden   inGroup   conductorIdx
```

- `n0`, `n1`: endpoint node indices (0-based)
- `maxSideLength`: mesh size constraint (double; -1 = auto)
- `propIdx`: boundary property index (1-based; 0 = none)
- `hidden`: 0 = visible, 1 = suppressed from mesh
- `inGroup`: group number
- `conductorIdx`: conductor index (1-based; 0 = none)

2.6.3 Arc Segments

```
[NumArcSegments] = <count>
n0  n1  arcAngle maxSideLength propIdx hidden inGroup conductorIdx mySideLength
```

- `n0`, `n1`: endpoint node indices (0-based)
- `arcAngle`: arc span (degrees)
- `maxSideLength`: mesh constraint (double)
- `propIdx`: boundary property index (1-based; 0 = none)
- `hidden`: 0 = visible, 1 = suppressed
- `inGroup`: group number
- `conductorIdx`: conductor index (1-based; 0 = none)
- `mySideLength`: computed side length (0 = auto)

2.6.4 Holes

```
[NumHoles] = <count>
x    y    inGroup
```

2.6.5 Block Labels

```
[NumBlockLabels] = <count>
x    y    blockIdx   meshSize   inGroup   extFlags
```

- `x`, `y`: label position (double)
- `blockIdx`: material index (1-based; 0 = unassigned)
- `meshSize`: element size (double; ≤ 0 = auto). Stored as diameter; converted to area: $A = \pi d^2/4$
- `inGroup`: group number
- `extFlags`: bit 0 = external region, bit 1 = default region

2.7 Differences from Magnetics and Electrostatics Formats

- `[dT]` and `[PrevSoln]` enable transient analysis (unique to heat flow).
- Material properties use thermal conductivity (K_x, K_y), volumetric heat capacity (K_t), and heat generation (q_v).
- Nonlinear materials use T-K curves (temperature vs. conductivity) instead of B-H curves.
- Boundary conditions include convection (h, T_∞) and radiation (β, $T_{\infty,\text{rad}}$), which are physics-specific to heat transfer.
- Point properties use temperature T_p and point heat source q_p.

3 The `.anh` Output File

The `.anh` file begins with a verbatim copy of the input `.feh` file (all sections described in Section 2), followed by the solver's solution data.

3.1 Solution Header

Immediately after the echoed `.feh` content:

```
[Solution]
```

3.2 Mesh Nodes

```
<NumNodes>
```

followed by `NumNodes` lines:

```
x    y    T    Q
```

- `x`, `y`: node coordinates in model units
- `T`: computed temperature (K)
- `Q`: boundary/conductor marker (int):
  - -2 = interior node (no boundary condition)
  - -1 = fixed temperature boundary
  - $0 \ldots N-1$ = conductor index

Node indices are 0-based throughout the solution section.

3.3 Mesh Elements

```
<NumElements>
```

followed by `NumElements` lines:

```
p0    p1    p2    labelIdx
```

- `p0`, `p1`, `p2`: vertex node indices (0-based)
- `labelIdx`: block label index (0-based, into the block label list from the `.feh` header)

3.4 Conductor Solution Data

```
<NumConductors>
```

followed by `NumConductors` lines:

```
T    q
```

- `T`: temperature on the conductor (K). For prescribed-temperature conductors, this is the prescribed value; for prescribed-flow conductors, this is the computed value.
- `q`: total heat flow on the conductor (W). For prescribed-flow conductors, this is the prescribed value; for prescribed-temperature conductors, this is the computed value.

4 Index Conventions

- **Property indices** (materials, boundaries, conductors, point properties) are **1-based** in the file. An index of 0 means "no property assigned." The parser converts to 0-based internally.
- **Node indices** (in segments, arcs, and the solution section) are **0-based**.

- **Block label indices** in the solution element list are **0-based**, referencing the block label array from the `.feh` header.

5 Unit Conversion

The `[LengthUnits]` tag determines the coordinate system. All coordinates in the file are in the specified units. Conversion to meters:

Unit	Meters per unit
inches	0.0254
millimeters	0.001
centimeters	0.01
meters	1.0
mils	2.54×10^{-5}
microns	10^{-6}